

4.4 Non-parametric tests

Further Probability and Statistics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Rank-based tests use order rather than a specified normal distribution. A sign test checks direction of paired changes after discarding zero differences; under null, plus signs follow $\text{Binomial}(n, 0.5)$. Wilcoxon signed-rank additionally uses magnitudes via ranks, subject to its assumptions.

Worked example: follow the reasoning

If 8 nonzero paired differences are equally likely positive/negative under H_0 , number positive $X \sim \text{Bin}(8, 0.5)$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 In six paired observations, five differences are positive and one negative. Give two-sided sign-test p-value under H_0 . [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$P(X \geq 5) = C(6,5)/64 + C(6,6)/64 = 7/64$ [2]; two-sided $p = 14/64 = 0.21875$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.